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import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical matrix o
import matplotlib.pyplot as plt # Import Matplotlib to visualize the out

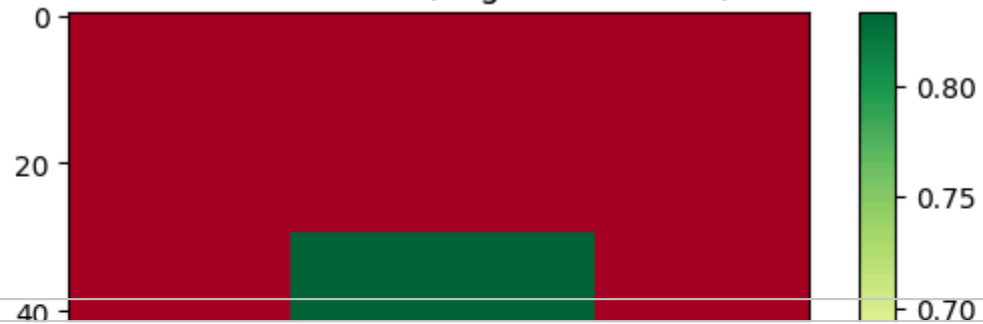
# Step 1: Simulate the acquisition of Red and Near-Infrared (NIR) satellite bands
red_band = np.full((100, 100), 50, dtype=np.float32) # Create a solid gray image array filled with a
nir_band = np.full((100, 100), 150, dtype=np.float32) # Create a solid gray image array filled with

# Step 2: Introduce a "forest" area in the center (High NIR reflection, Low Red reflection)
cv2.rectangle(red_band, (30, 30), (70, 70), 20, -1) # Draw a filled or outlined rectangle on the ima
cv2.rectangle(nir_band, (30, 30), (70, 70), 220, -1) # Draw a filled or outlined rectangle on the im

# Step 3: Calculate the Normalized Difference Vegetation Index (NDVI)
# Formula: (NIR - Red) / (NIR + Red)
ndvi = (nir_band - red_band) / (nir_band + red_band) # Calculate Normalized Difference Vegetation In

# Step 4: Display the vegetation map
plt.imshow(ndvi, cmap='RdYlGn') # Render the image array to the display
plt.title('Calculated NDVI (Vegetation Index)') # Add a descriptive title to the plot
plt.colorbar() # Add a color legend bar to the plot
plt.show() # Show the final plot window to the user
```

Calculated NDVI (Vegetation Index)



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